

BMJ Open Dengue Reduction through Education Awareness and Monitoring in Schools (DREAMS) initiative in Puducherry, India: protocol for a hybrid type 3 effectiveness-implementation study

Arya Rahul ¹, Srikanth Srirama,¹ Jency Priskilla ¹,
Gnanasekaran Vijayalakshmi,¹ Amala Ramasamy,¹
Ananganallur Nagarajan Shriram,¹ Vasantha Kumari Ramadass,²
Sivasankari Thirumurthy,³ Manju Rahi ¹

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¹ICMR - National Institute of Vector Control Research, Puducherry, India

²National Vector Borne Disease Control Programme, Directorate of Health and Family Welfare Services, Puducherry, India

³Department of School Education, State Training Centre, Puducherry, India

Correspondence to
Dr Jency Priskilla;
jjpriskilla@gmail.com

ABSTRACT

Background Dengue continues to be a global public health challenge despite sustained control efforts. Dengue control strategies rely primarily on effective *Aedes* mosquito control. Sustaining community participation in mosquito control requires innovative efforts, especially in achieving consistent behavioural change. Schools offer a strategic entry point, with children being both learners and influencers. This leads to a ripple effect of intervention impacting beyond the classroom. While behaviour change strategies for dengue prevention exist, there is sparsity of studies addressing the implementation aspects including contextual factors and sustainable mechanisms.

Methods and analysis We propose a hybrid type 3 effectiveness-implementation study to design, implement and evaluate a school-based dengue control initiative in Puducherry. The study will be conducted in three phases: (1) formative research and co-development of the intervention package and implementation strategies, (2) implementation and (3) evaluation. The intervention will be co-developed using the principles of Communication for Behavioural Impact, with school teachers serving as key change agents to deliver knowledge and promote dengue prevention practices among students. The Consolidated Framework for Implementation Research 2.0 will guide the implementation process across all phases. Evaluation will be carried out using the Reach, Effectiveness, Adoption, Implementation and Maintenance framework, adopting a mixed-methods approach to assess intervention effectiveness, implementation outcomes and cost implications.

Ethics and dissemination The study protocol was approved by ICMR- NIVCR Institutional Human Ethics Committee (IHEC 0123/N/D). All standard ethical practices will be adhered to. Written informed consent (or age-appropriate assent) will be obtained from all participants. Findings will be shared via policy briefs/ reports to stakeholders followed by publication in reputed journals.

Trial registration number CTRI/2026/02/104985

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ This protocol addresses a key gap in implementation research on school-based dengue interventions by applying a hybrid type 3 effectiveness-implementation design, enabling simultaneous assessment of implementation outcomes and effectiveness of a contextually grounded, theory-informed initiative in Puducherry. This study aims to offer a scalable model for implementation-driven vector control in low-resource settings.
- ⇒ The intervention will be co-developed with stakeholders across the health, education and local governance sectors, ensuring intersectoral collaboration and contextual fit. Novel tools for engaging and monitoring participants will be co-developed as part of the study to facilitate real-time feedback and adaptive implementation.
- ⇒ The study will apply multiple implementation frameworks—Consolidated Framework for Implementation Research 2.0 and Behaviour Change Wheel for developing the implementation strategy and Reach, Effectiveness, Adoption, Implementation and Maintenance for evaluation.
- ⇒ The lack of a control group limits causal inference as changes cannot be attributed solely to the intervention. The pre-post design was a pragmatic choice given the policy decision to implement the intervention across all schools and the fixed academic calendar.
- ⇒ The effectiveness assessment will primarily rely on self-reported behaviours and routine surveillance data, which may not fully capture real-time changes in vector indices or dengue transmission.

BACKGROUND

Dengue has emerged as the most significant mosquitoborne viral disease globally, with a 10-fold increase in cases over the past two decades.¹ Approximately half of the world's

population is now at risk of dengue, and this number is expected to rise further due to rapid urbanisation, increased travel and climate change.^{2 3} India is one of the largest contributors to the global dengue burden, reporting over 200 000 cases annually. However, considering under-reporting and underdiagnosis, the actual burden could be up to 30 times higher, highlighting the need for enhanced surveillance and more comprehensive disease burden assessments.^{4–6} Despite the scale of the problem, sustained, scalable solutions remain limited.

The National Vector Borne Disease Control Programme (NVBDCP), under the Ministry of Health and Family Welfare, serves as the national framework for the prevention and control of vectorborne diseases, including dengue in India.⁷ The dengue control strategies remain largely dependent on vector control and outbreak response, both of which are difficult to sustain in routine practice. This highlights the need for interventions that are not only effective but also feasible, scalable and embedded within existing systems.⁸

Challenges in dengue prevention and control

Given the absence of a specific treatment for dengue and the scepticism surrounding dengue vaccines, implementing effective vector control measures appears to be a crucial tool in curbing the transmission of dengue.⁹ The primary strategies used in India include the use of chemical larvicides such as temephos (Abate) in breeding containers, environmental management, source reduction and community awareness campaigns.⁷ Meta-reviews indicate variable effectiveness of these strategies: chemical interventions (eg, Indoor Residual Spraying (IRS)/space spraying) achieve 50%–90% transient adult reductions but rebound within 4–7 days due to *Aedes* resting behaviours and resistance. Additionally, the effectiveness of these approaches has certain limitations. First, insecticide resistance among *Aedes aegypti* and *Aedes albopictus* has reduced the impact of chemical-based interventions.¹⁰ Second, the cost and logistical challenges of sustained vector control programmes pose a significant barrier to long-term success. Third, community engagement remains weak, as many communities perceive dengue as a mild illness, leading to poor adherence to prevention practices.^{11–14} Findings from previous studies in the setting highlight critical gaps in dengue prevention, including low perceived susceptibility, limited awareness of mosquito breeding sites and an overreliance on government-led initiatives.¹¹ A recent capacity-building initiative conducted in Puducherry demonstrated the feasibility of structured training programmes to improve knowledge and engagement in dengue prevention among students and multidisciplinary staff in medical college settings.¹⁵ However, persistent behavioural and systemic challenges were noted, emphasising that training alone is insufficient without complementary community-based strategies. Hence, the key gap is not simply knowledge deficiency, but the absence of an implementation strategy that can convert school-based awareness into

routine, sustained action across households, schools and local health systems. Certain innovative entomological mosquito control methods offer a promising solution, but needs to be extensively evaluated in Indian settings before roll-out.¹⁶

These insights emphasise the need for tailored implementation strategies that address local knowledge gaps and behavioural barriers. Moreover, studies have shown that knowledge alone is insufficient to drive behaviour change; sociocultural and demographic factors play a significant role in influencing preventive behaviours.^{11 17–19} The repeated failure of large-scale vector control programmes highlights the need for more sustainable, community-focused strategies.²⁰ Strengthening behavioural change and community mobilisation is critical to improving the adoption and sustainability of dengue prevention measures.^{21 22} This calls for a shift from top-down, information-driven approaches to more participatory, community-based strategies.

Implementation research approaches in dengue control

Implementation research (IR) offers a structured framework to identify barriers in intervention delivery, adapt strategies to local contexts and ensure sustainable implementation.²³ In the context of dengue control, IR is essential for assessing the real-world effectiveness of vector control programmes and refining strategies to enhance community engagement and behaviour change. Despite numerous community-based interventions aimed at dengue prevention, significant gaps remain in their implementation and long-term sustainability.

Various participatory approaches have been employed to strengthen community engagement in dengue control, including mobilising family leaders,²⁴ volunteer groups,²⁵ school students^{26–28} and front-line health workers.²⁹ Additionally, digital platforms such as mobile applications, web-based tools and real-time notification systems have been leveraged to improve participation and response to dengue outbreaks.^{30–32} However, the majority of these studies rely on quasi-experimental or controlled trial designs and predominantly evaluate short-term outcomes such as changes in knowledge, entomological indices or vector breeding site reductions.

A critical research gap exists in the structured application of IR to assess key outcomes such as acceptability, feasibility, sustainability and cost-effectiveness of dengue prevention strategies. A health system evaluation of the study setting reveals that, despite efforts to raise awareness about dengue in schools, these activities are severely hindered by weak intersectoral collaboration.³³ The success of community-based interventions relies on their adaptability to local contexts, community ownership and political commitment.^{34–36} However, limited studies have explored stakeholder-driven co-development of vector control interventions, assessed their transferability across diverse settings or examined their integration into public health frameworks. Beyond evaluating intervention effectiveness, a stronger focus is needed on participatory

approaches that align with socio-political dynamics and facilitate the institutionalisation of dengue control strategies within health and governance systems. The main gap is not a lack of dengue awareness interventions, but the absence of an implementation strategy that can embed school-based prevention into routine health and education systems in a sustainable way.

Empowering children in dengue prevention efforts

Children and adolescents are strategically positioned to drive community-wide behaviour change for dengue prevention. Studies have shown that they can act as agents of change, transferring knowledge from the classroom to their families and communities.³⁷ By involving children in dengue prevention programmes, schools can create a ripple effect of behaviour change at the household and community levels.²⁷ Programmes in Southeast Asia and Latin America have shown that children who receive structured dengue education are more likely to adopt mosquito control practices at home and motivate their families to participate in community-based prevention activities.^{38 39} However, most school-based interventions have focused primarily on improving knowledge rather than sustaining behavioural change.

The Dengue Reduction through Education Awareness and Monitoring in Schools (DREAMS) addresses this gap by combining formative co-design, school-based delivery through teachers, and monitoring within a real-world implementation framework. It leverages Wenger's Community of Practice (CoP) concept, fostering peer learning and shared experiences among students, teachers, and parents.⁴⁰ Its novelty lies in linking dengue prevention to existing school systems while simultaneously studying both intervention delivery and implementation outcomes. It aims to co-develop an intervention with systemic alignment across the health and education sector, integrating theoretical frameworks. Implemented as a tripartite initiative involving researchers, health officials and educators, this study will assess the intervention's effectiveness along with the implementation outcomes. By embedding dengue prevention into curricula and fostering teacher-led engagement, DREAMS can help transform schools into active hubs for long-term vector control. Teachers can serve as long-term facilitators, as they can integrate dengue prevention messages into regular curricula and reinforce behavioural change over time. The findings could have broader implications for vectorborne disease control and other health education strategies.

Objectives

This study aims to co-design, implement and evaluate a school-based intervention for dengue control in Puducherry, through teachers as key delivery agents. The study will develop a standardised training module and digital tools through formative research and assess implementation outcomes such as feasibility, acceptability,

fidelity and cost, alongside preliminary behavioural outcomes.

METHODOLOGY

This project involves intersectoral research integrating health policy and systems research with implementation science, behavioural change communication and digital tools through a hybrid type 3 implementation—effectiveness design. A hybrid type 3 design is appropriate because the primary emphasis is on refining and testing the implementation strategy while also generating preliminary evidence on the intervention's effectiveness in routine settings. This study protocol has been reported in accordance with the TIDieR (Template for Intervention Description and Replication) checklist⁴¹ provided as online supplemental additional file 1. A multilevel participatory approach will be used, engaging key stakeholders from multiple sectors to co-design the intervention and its implementation strategies. This approach will be integrated into every project phase, from shaping the research framework to implementing the intervention and analysing and disseminating the findings. School teachers, designated as 'DREAMS ambassadors', will deliver the intervention, engaging students and providing feedback to refine and adapt the implementation strategy for future use. A conceptual workflow of the planned intervention is shown in figure 1.

Considering the complexity of the intervention as per elements of emergence, feedback, adaptation and self-organisation described by Skivington *et al*, priority will be given to design, implement and evaluate the intervention to contribute comprehensively for a system change.⁴² The Consolidated Framework for Implementation Research (CFIR) 2.0 will be applied throughout all stages of the study to ensure a comprehensive understanding of the implementation process.⁴³ CFIR 2.0 will guide the identification of contextual barriers and facilitators, inform the choice and adaptation of strategies to align with local needs and help assess implementation outcomes, supporting continuous improvement. Prior identification of key constructs will allow researchers to address factors influencing successful implementation.⁴⁴

Study setting

This study is planned in Puducherry, the capital of the Puducherry Union Territory in India, which stretches 294 sq. km along the Bay of Bengal's Coromandel Coast and shares borders with Tamil Nadu state.⁴⁵ The region is known for its French colonial heritage, dense population and diversity of urban, semiurban and rural communities. The tropical climate of Puducherry, characterised by elevated humidity, warm temperatures and seasonal monsoon rains, creates ideal conditions for mosquito breeding and the spread of *Aedes* borne diseases like dengue. Additionally, urbanisation and population density contribute to artificial breeding sites, such as water storage containers and discarded waste,

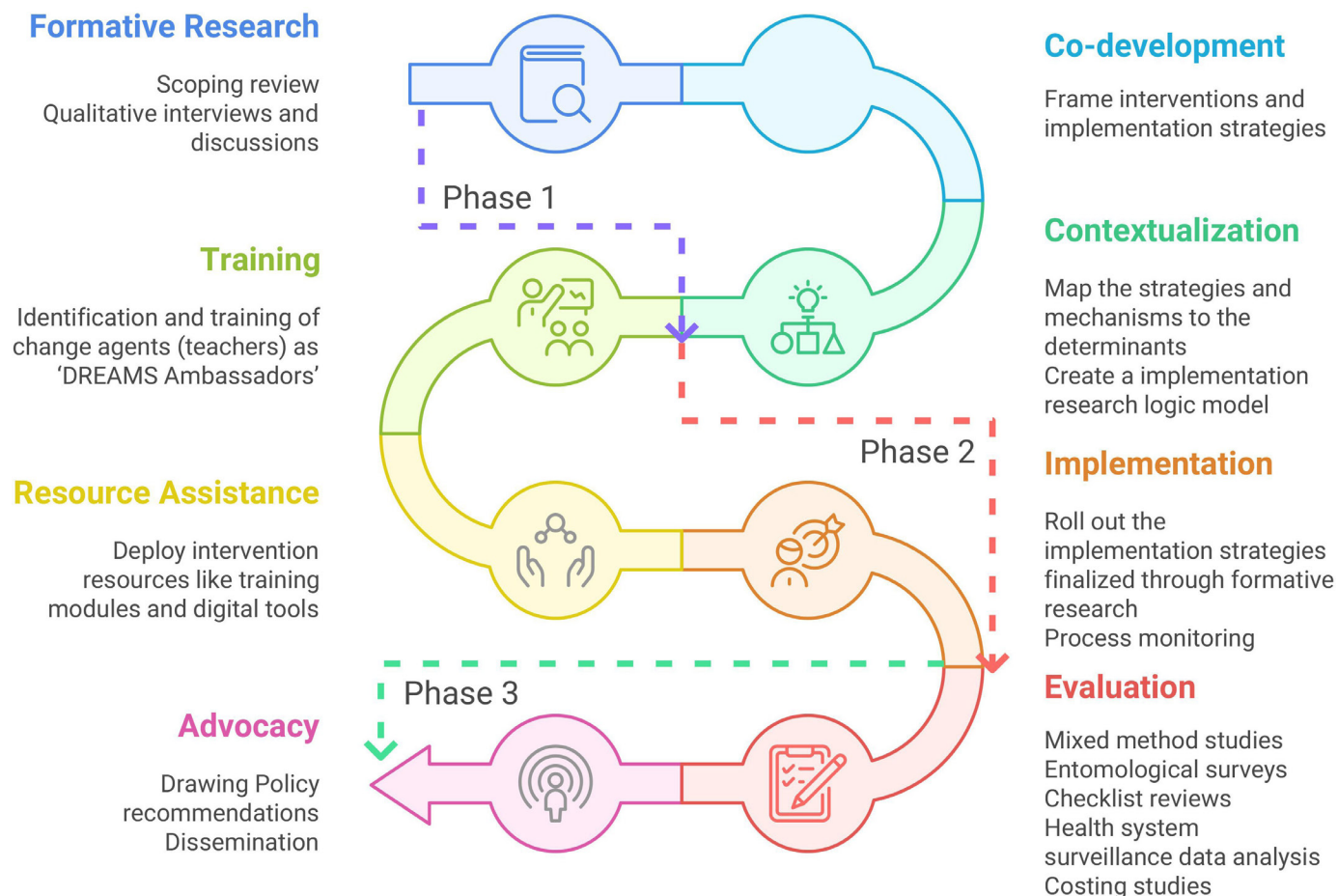


Figure 1 Key activities proposed under Dengue Reduction through Education Awareness and Monitoring in Schools (DREAMS) initiative.

further facilitating the proliferation of *Aedes* mosquitoes. Following a significant dengue outbreak in 2017, which reported more than 4000 cases, the area has had recurrent outbreaks with co-circulation of all four virus serotypes.⁴⁶ Though peak dengue incidence is reported from October to January, cases are reported throughout the year, raising concerns regarding continuous prevention and control activities.

Study procedure

The study will be conducted in three phases. In the preimplementation phase, a formative evaluation will be carried out to co-develop tailored implementation strategies, considering contextual facilitators and barriers. The implementation phase will then deploy these strategies across all schools in Puducherry district, with ongoing monitoring. Finally, the evaluation phase will assess the intervention's impact and provide policy recommendations for sustainable school-based vector control interventions.

In phase I, CFIR 2.0^{43 47} will guide assessment of contextual barriers and facilitators, while Behaviour Change Wheel (BCW) and Expert Recommendations for Implementing Change (ERIC) will support translation of identified determinants into implementation strategies. In phase II, Communication for Behaviour Impact (COMBI)

will inform the content and delivery of the behaviour-change intervention. In phase III, Reach, Effectiveness, Adoption, Implementation and Maintenance (RE-AIM) and CFIR 2.0 will be used to assess implementation outcomes, interpret contextual influences and explain variation in adoption, fidelity and maintenance. This framework integration will be organised through a logic model to ensure coherence and avoid overlap. An overview of all the theories/frameworks used in each phase is given as a table in online supplemental additional file 1. The study will use the frameworks in a sequential and complementary manner across the three phases.

Phase 1: formative research and co-development of key implementation strategies

The formative research will be guided by the CFIR 2.0 model, which provides a structured framework for analysing and addressing contextual factors that impact the implementation process. The CFIR framework encompasses five key domains: intervention characteristics, outer setting, inner setting, characteristics of individuals and process. These domains will help identify facilitators and barriers, enabling the development of targeted strategies to enhance the intervention's feasibility, acceptability and scalability.

As a preliminary step, a scoping literature review on similar health interventions in school settings will be conducted to refine the implementation approach. Additionally, stakeholder consultation meetings will be held to explore internal and external contextual factors. Policy documents, peer experiences, health system reports and school-based health programme records will be analysed to identify gaps and opportunities for school-based dengue prevention interventions. These insights will inform the development of a logic model, mapping the relationships between inputs, activities, outputs and expected outcomes.⁴⁸

To ensure a comprehensive understanding of implementation barriers and facilitators, key stakeholders from both the education and health sectors will be identified and engaged. This will include teachers, students and administrators from the education sector and health workers, entomologists and programme officers from the health sector. Stakeholder engagement will be conducted through structured interviews, focus group discussions (FGDs) and consultative meetings. These interviews will be guided by the BCW framework to assess behavioural determinants and identify feasible interventions suited to the target population. By integrating BCW within the theoretical framework, the study will conduct a causal analysis of behavioural targets to inform intervention design.⁴⁹

These formative sessions will serve a dual purpose: (1) to develop intervention strategies based on BCW principles and (2) to capture contextual barriers and facilitators in alignment with CFIR domains. Engaging stakeholders from the outset helps ensure that the intervention is grounded in local needs, priorities, and realities, making both its design and implementation more relevant and effective. Insights from these sessions will also be used to refine the logic model, ensuring it accurately reflects stakeholder perspectives and formative research findings.

Building on the insights from the formative phase, tailored implementation strategies will be co-developed with key stakeholders from the education and health sectors. The behavioural determinants and intervention components best suited for the school settings will be identified and the selected strategies will be refined using ERIC guidelines to ensure feasibility, adaptability and sustainability. The primary goal of these strategies will be to empower teachers with the knowledge and skills needed to actively involve students in dengue control efforts, fostering sustained behavioural change. To support this, standardised educational and digital tools will be developed using COMBI principles, enhancing student engagement and promoting community-driven vector control initiatives. These resources will strengthen teachers' ability to deliver effective health education, increasing awareness and encouraging lasting behavioural shifts among students. Additionally, they will serve as a scalable model for expanding school-based vector control programmes in other settings.

The intervention will be finalised through the formative phase results in which content, delivery format, duration, frequency and fidelity procedures will be refined in consultation with stakeholders to ensure contextual fit and feasibility. Accordingly, the protocol presents the planned intervention at a conceptual level, while the detailed operational specifications will be finalised after phase I and documented in the TIDieR checklist (online supplemental additional file 2).

Phase 2: implementation of the school-based intervention

The implementation phase will involve ongoing stakeholder engagement, fostering collaboration across departments such as health and school education, with shared ownership of the programme. Prior to rolling out the intervention, baseline assessments will be conducted among teachers, students and parents along with entomological surveys to record the preintervention indices. Central to the intervention is the training of teachers, who will act as change agents and DREAMS Ambassadors. Educators from schools across Puducherry district will participate in structured training sessions focusing on knowledge dissemination, hands-on demonstrations and familiarisation with the tutor's guide. The trained teachers will then deliver the intervention to students using a structured module that incorporates weekly and monthly activities, community engagement efforts and experiential learning approaches. All these activities will be embedded within routine school activities.

To ensure sustained engagement, continuous monitoring and feedback mechanisms will be established. School-based digital and educational tools will be assessed for usability and effectiveness in reinforcing behavioural change. Ongoing interactions with stakeholders will help refine the intervention, address emerging challenges and strengthen integration within the school system.

Phase 3: evaluation

Outcome measurement will combine self-reported behavioural indicators with objective entomological and routine surveillance measures to reduce dependence on any single source of evidence. To minimise recall and social desirability bias, all questionnaires will be administered anonymously using standardised instructions, and respondents will be assured that participation will not affect any school or health service-related benefits. Survey instruments will undergo pilot testing at the outset of the formative phase to assess comprehensibility, contextual relevance and preliminary psychometric performance, after which necessary modifications will be made before main study implementation. The relationship between entomological indicators and dengue incidence will be interpreted cautiously and examined within the broader analytical framework as a supportive rather than sole proxy of transmission dynamics.

The evaluation will be guided by the RE-AIM framework to measure the programme impact, while CFIR 2.0 will provide insights into implementation processes and

Table 1 Overview of evaluation plan

Outcome indicators	Measures/tools	Source/participants	Timing of measures
Effectiveness outcomes			
Primary			
Change in knowledge, perceptions and behaviour	Quantitative survey	Students, parents, teachers	Preintervention and postintervention
Secondary			
Change in adult <i>Aedes</i> density in schools	Gravid trap mosquito collections	In school campus	Weekly monitoring preintervention and during intervention
Change in the presence of <i>Aedes</i> breeding sources in schools	Breeding source audits	In school campus	Monthly monitoring preintervention and during intervention
Dengue incidence	Routine surveillance reports	Secondary data	5 years preintervention and during the intervention periods
Larval indices in the community	Routine surveillance reports	Secondary data	3 years preintervention and during the intervention periods
Implementation outcomes			
Outcome indicators	Measures/tools	Source/participants	Timing of measures
Reach: Extent of engagement of parents and community	1. Reporting through digital tool	Students Parents	Postintervention implementation
Adoption: Extent of engagement of teachers and schools	2. Quantitative survey 3. Checklists 4. Qualitative interviews (IDIs, FGDs)	► Teachers ► School administration ► Officials from education department	
Implementation: ► Acceptability ► Appropriateness ► Feasibility ► Fidelity Barriers and facilitators	► Quantitative surveys (IAM, FIM) ► Qualitative interviews (IDIs and FGDs) Checklists	Students, parents, teachers	
Maintenance: Continuation of activities beyond the intervention period	Observations, digital tool reporting, qualitative interviews	Teachers, students, school administration, officials from health and education departments	At 1 year and 3 year postintervention
Cost: Cost per school trained, cost per student educated, cost per unit improvement in knowledge and behaviour scores and cost per unit reduction in vector indices	Budget records, administrative reports, and stakeholder interviews	School administration, teachers, officials from health and education departments	Postintervention implementation
FGDs, focus group discussions; FIM, Feasibility of Intervention Measure; IAM, Intervention Appropriateness Measure; IDIs, in-depth interviews.			

outcomes.⁴³ A mixed-methods approach, combining quantitative and qualitative techniques, will be used to gain a well-rounded understanding of the project's performance and overall impact. Additionally, exit interviews with DREAMS ambassadors will be conducted to explore barriers and facilitators to implementation, guided by CFIR 2.0. These interviews may also capture unintended effects, areas for improvement and recommendations for scaling up the intervention to a broader geographic area. [Table 1](#) presents a summary of the planned evaluation approach.

Reach

Reach will be measured by the proportion and demographics of students who participate in the programme and compared with the total eligible population in participating schools. Quantitative data will include student enrolment rates in dengue prevention activities and engagement levels captured through the digital tool. Demographic data will be used to assess equitable reach across various groups. Additionally, surveys and FGDs with students and parents will explore factors influencing participation, including awareness, perceived relevance and barriers to engagement.

Effectiveness

The effectiveness of the intervention will be assessed through school-based, clustered pre-post quantitative surveys. Twenty schools will be selected from the ten highest dengue-endemic areas of Puducherry. Because the intervention is delivered at the school level, unequal cluster sizes are anticipated, and the sample size calculation accounted for clustering, with students nested within schools. The required sample size was estimated using standard formulas for comparing mean preintervention and postintervention knowledge scores, assuming a 5% level of significance, 80% power, an SD of 12 units, an average cluster size of 365 students per school, a cluster size variance of 75, an intracluster correlation coefficient of 0.45, and a minimum detectable difference of 2 units in knowledge, attitudes and practice (KAP) scores. After adjusting for clustering, the required sample was 3141 students. Further allowance for 20% non-response and 10% attrition yielded a final sample of 4363 students across 20 schools.

The ten high-endemic areas will be identified using National Centre for Vector Borne Disease Control (NCVBDC) surveillance data based on cumulative dengue incidence over the preceding 3 years. Areas will be ranked by incidence per 1000 population, and the 10 areas with consistently high case burdens will be included. Within these areas, a proportional stratified sampling approach will be used to ensure representation of both government and private schools and to minimise selection bias. Changes in KAPs among students and parents will be measured using validated structured questionnaires administered at baseline and again after 6 months. Similar preintervention and postintervention assessments will be conducted among teachers designated as dengue ambassadors to evaluate changes in their KAPs.

Eligible participants include grade 6–8 students (ages 11–14 years) enrolled in selected schools and present on survey days, their primary caregivers able to complete surveys in Tamil or English, and teachers designated as DREAMS ambassadors (1–2 per school) with minimum 1 year teaching experience and willing to participate in training.

The effectiveness of vector control measures adopted within the schools will be evaluated by assessing the entomological indicators such as *Aedes* density per trap and larval indices. *Aedes* gravid traps will be placed in the 20 selected schools, and *Aedes* density per trap will be monitored weekly to evaluate changes in mosquito populations. Monthly mosquito breeding site audits will be conducted in the selected schools to assess the implementation of dry-day activities. Household-level *Aedes* breeding site audits will be conducted through survey reports submitted by volunteering students through the digital tool, with independent validation by the NVBDCP team for accuracy and completeness.

Data from routine vector surveillance conducted by the NVBDCP will be used to evaluate the broader community-level impact of the intervention. Trends in

dengue incidence, based on data from the Integrated Disease Surveillance Programme, will also be analysed to identify shifts in disease patterns and assess the overall impact of the programme. Programme impact on dengue-related health outcomes of children will be monitored through fever surveillance and school absenteeism reports received from the participating schools. Real-time reporting and response mechanisms will be established to avoid potential dengue outbreaks in the study area.

Adoption

Adoption will be assessed by measuring the extent to which schools and teachers engage with programme activities. Quantitative adoption metrics will include the number and proportion of invited schools that implement the programme, the number of teachers trained. Digital tools will be used to track activity reporting. Additionally, qualitative interviews or surveys with school administrators and teachers will explore factors influencing adoption, including perceived barriers and facilitators. This assessment will provide insights into the programme's actual penetration across diverse educational settings. Participation in training sessions and the completion of programme activities will be tracked, ensuring accurate and timely reporting.

Implementation evaluation

Implementation outcomes will be evaluated by mapping the CFIR 2.0 constructs and sub constructs. The intervention characteristics domain will assess the programme's complexity and adaptability, while the outer setting domain will examine factors like NVBDCP support and community acceptance. The inner setting domain will explore school-level leadership, resource availability and teacher engagement. The 'characteristics of individuals' domain will focus on teacher motivation and student engagement, and the process domain will evaluate the planning, execution and adaptation of programme components over time.

Organisational readiness, including teacher preparedness and school-level support, will be assessed through qualitative interviews and surveys. The external environment, including NVBDCP support and government collaboration, will also be analysed using qualitative tools to understand the broader policy and infrastructural context. Intervention characteristics, such as acceptability, feasibility and appropriateness will be evaluated through teacher and student feedback surveys, interviews and discussions. Acceptability of Intervention Measure, Feasibility of Intervention Measure and Intervention Appropriateness Measure will be used to prepare the quantitative survey tools. Implementation fidelity will be tracked through real-time monitoring of activity completion, and periodic review meetings with stakeholders and analysed using a checklist.

Barriers and facilitators to the implementation of the programme will be assessed from an organisational and individual perspective through KII, in-depth interview

(IDI) and FGDs with different levels of stakeholders at various stages of the intervention. The interview guides will be framed using the logic model developed. The BCW will be used to understand the effect of intervention in the participants. The relationships between various stakeholders and how it facilitated the programme implementation will be explored to understand the facilitating networks within the system, for future adaptations.

Maintenance

Maintenance potential of the programme will be captured in surveys and qualitative interviews. It will be further evaluated by examining the continuation of programme activities beyond the intervention period and analysing the long-term trends in dengue incidence and programme participation.

Cost

Costing will be done from the perspective of the system (implementing agency) using a hybrid approach that combines Ingredients-Based Costing and Time-Driven Activity-Based Costing (TDABC) methodologies.⁵⁰ Fixed costs will cover one-time expenses such as the development of educational materials, training of teachers and procurement of materials. Recurring costs will include expenses for holding meetings, incentivising, monitoring and evaluation as well as programme implementation costs like logistics. Cost items will be classified by resource type, key activity and phase of implementation (design, initiation and maintenance), consistent with established implementation costing frameworks.⁵¹ TDABC will estimate personnel costs by mapping activities, assigning time estimates and multiplying them by unit costs, ensuring precise labour cost estimation. Data on these expenditures will be collected by mixed methods from budget records, administrative reports and stakeholder interviews. All costs will be reported in INR at 2024–2025 reference year prices. Given the short programme implementation duration, discounting will not be applied, consistent with standard practice for short-term programme evaluations.

Since the universal school-based rollout design of DREAMS precluded the formation of a concurrent control group for operational and ethical reasons, traditional incremental cost-effectiveness ratios (ICERs) cannot be calculated. Alternatively, pragmatic cost-effectiveness metrics will be adopted in lieu of ICERs, consistent with the flexibility permitted under the iDSI Reference Case for Economic Evaluation in Low and Middle Income Countries (LMICs).⁵² The preintervention survey will serve as the baseline reference state, providing a within-group temporal comparator for all outcome measures. Effectiveness will be measured through pre–post surveys considering outcomes like process indicators, behavioural change outcomes, vector indices and health indicators. Cost per school trained, cost per student educated, cost per unit improvement in knowledge and behaviour scores and cost per unit reduction in vector indices will be calculated. Additionally, historical dengue case data

will be examined to understand broader epidemiological trends, though direct attribution to the intervention will remain a challenge. One-way sensitivity analysis will evaluate the impact of $\pm 20\%$ variation in key cost drivers (personnel costs, material costs and participation rates) on programme feasibility and cost-effectiveness estimates. This pragmatic cost assessment is expected to offer insights into the financial feasibility and scalability of DREAMS in similar educational and public health settings.

Data collection and analysis

Data collection will involve both quantitative and qualitative methods. The behavioural survey will be administered anonymously, with standardised instructions emphasising that there are no right or wrong answers, to reduce social desirability pressure. Self-reported indicators will be triangulated with objective entomological measures and routine surveillance data, allowing comparison across data sources rather than relying on a single outcome domain. The survey instruments will be pre-tested for validity (content and construct), cultural appropriateness, acceptability and reliability before full-scale administration. Internal consistency will be measured using Cronbach's alpha. Digital tools, questionnaire surveys, entomological surveys and routine surveillance reports will provide quantitative data on programme reach, adoption and effectiveness. In addition, the wording and sequencing of items will be refined based on stakeholder feedback to minimise misunderstanding and response bias. Structured questionnaires for pretraining and post-training assessments will be. Demographic data such as age, gender, urban–rural settings and school type will be collected to assess representativeness. Descriptive statistics will be used to summarise participation rates, knowledge, perspectives and behaviour improvements and entomological outcomes.

To evaluate the effectiveness of the intervention, linear mixed-effects model (LMM) will be used to compare post-intervention KAP scores among students, teachers and parents while adjusting for baseline differences. Baseline (preintervention) scores corresponding to each outcome variable will be selected as covariates in the respective LM models. Specifically, preknowledge score will be used as a covariate for postknowledge score, preattitude score for postattitude score and prepractice score for postpractice score. These covariates will be selected based on Wald test or Likelihood ratio tests and the various competing models will be compared using Akaike and Bayesian Information Criteria. Model fitness to the data will be verified using model diagnostics statistics and visually by residual plots. If significance in the result is observed, post hoc pairwise comparisons between groups will be performed on the estimated marginal means obtained from the LMMs, using Tukey-adjusted tests. A $p < 0.05$ was considered statistically significant. Missing data will be assessed for extent and pattern before analysis. When the proportion of missing data is minimal ($< 5\%$), complete case analysis will be performed. For variables with higher

levels of missingness, appropriate imputation techniques will be considered to minimise bias. To control for type I error due to multiple comparisons, post hoc pairwise comparisons between groups were performed on the estimated marginal means obtained from LMM, using Tukey-adjusted tests to control the family-wise error rate. Equity will be assessed by comparing engagement levels across urban and rural schools, as well as among private, government and aided institutions. Stratified analysis will identify disparities in participation, implementation and programme effectiveness across these subgroups. A coding scheme will be developed for the open-ended questions, as far as possible, to enable a descriptive analysis. Otherwise, the data will be used for thematic analysis along with the qualitative data collected.

Qualitative data will be collected through IDIs and FGDs with teachers, students and parents purposively selected from relevant stakeholder groups to capture variation in experience, role and engagement with the intervention. Sampling will continue until data saturation is observed. All interviews and FGDs will be audio recorded with consent, and the recordings will be transcribed. The transcripts will undergo thematic analysis (using the approach outlined by Braun and Clarke),⁵³ with both deductive coding guided by CFIR constructs and inductive coding to capture emergent themes beyond the framework. Two investigators will independently code a subset of transcripts, compare coding and resolve discrepancies through discussion to enhance analytic consistency. Data triangulation across participant groups and data sources will be used to compare perspectives, identify convergence and divergence and strengthen credibility of the findings. Methodological rigour will be further supported through iterative codebook development, maintenance of an audit trail and regular team discussions to refine interpretations and ensure reflexivity throughout the analysis process.

All data collection instruments, including questionnaires and interview guides, will be translated into Tamil and piloted with 20–30 participants to refine language and content based on participant feedback. Adjustments will be made based on pilot findings to improve clarity and relevance. The data will be analysed using IBM SPSS Statistics V.22, STATA V.18 and R programming Language V.4.2.2.

Reporting and feedback

Real-time reporting through the digital tool will allow for continuous monitoring and rapid response to implementation challenges. Feedback loops will be established through peer group reminders, periodic review meetings with NVBDCP and school administrators and teacher feedback sessions. Findings from the evaluation will be disseminated to policymakers, health officials and participating schools to inform future dengue control strategies and support evidence-based improvements in programme delivery.

ETHICS AND DISSEMINATION

The study has received ethical clearance from the Institutional Human Ethics Committee of ICMR National Institute for Vector Control Research (IHEC 0123/N/D). All research activities adhere to ICMR ethical guidelines. Prior to data collection, a detailed participant information sheet explaining purpose, procedures, voluntary nature and right to withdraw without repercussions will be provided to all participants/legally authorised representatives (LAR—for minor participants). Written informed consent will be obtained individually from adult participants >18 years prior to participation. Parental/LAR consent and individual child assent will be secured using age-appropriate (written/verbal) assent forms explaining study participation in simple language. All interview sessions will be conducted by trained researchers. Participant confidentiality will be maintained through unique alphanumeric coding systems. No personal identifiers will appear in datasets or reports. Digital data will be password-protected and stored on encrypted institutional database with restricted access. Interviews will be audio recorded after separate consent and will be transcribed verbatim. Paper records will be stored in locked cabinets accessible only to core research team. The research findings will be communicated to stakeholders in the government of Puducherry for necessary action. They will also be published in peer-reviewed well-reputed journals.

DISCUSSION

This study adopts a hybrid type 3 effectiveness and IR design, ensuring to assess the impact and implementation outcomes within real-world settings. It follows a phased approach, beginning with formative research, progressing through co-design, implementation, monitoring and evaluation and ultimately aiming to inform policy recommendations for school-based vector control. DREAMS is a multidisciplinary stakeholder-driven initiative integrating health, school education and community engagement research for sustainable vector control practices in schools. Integrating participatory health research with implementation science provides a robust framework for effectively understanding and facilitating implementation processes. This integration supports the development of evidence-based yet contextually relevant interventions, enhancing their acceptability and scalability within institutional frameworks.⁵⁴ For example, combining Normalisation Process Theory with Participatory Learning and Action methodologies has been shown to strengthen implementation efforts by engaging stakeholders and embedding interventions into practice.⁵⁴ Additionally, the evolution of frameworks like integrated Promoting Action on Research Implementation in Health Services (i-PARIHS) underscores the importance of incorporating stakeholder perspectives to optimise implementation outcomes.⁵⁵

Studies have highlighted the importance of integrating interventions into existing workflows and organisational

structures.⁴¹ This is particularly crucial for programmes involving multiple departments, as embedding the intervention within the system enhances sustainability. Collaborative partnerships will help build capacity to sustain project goals beyond the funding period and ensure sustainability of the intervention.⁵⁶ Stakeholder consultations for the DREAMS initiative led to a consensus that implementation should cover all schools in Puducherry, given the recognised need and benefits. This ruled out a cluster-randomised design, as stakeholders opposed leaving schools as controls. A stepped-wedge design was considered but deemed impractical due to the academic calendar, where exams from January to March restrict intervention periods. The optimal window from August to December made phased rollout unfeasible. Instead, a pragmatic approach was adopted to ensure smooth implementation, impact and long-term adoption.

DREAMS aligns with Wenger's CoP framework by peer learning and shared identity in dengue prevention by engaging students, teachers and parents in collective knowledge-building.⁴⁰ Through structured peer interactions, experiential learning and mentorship, DREAMS cultivates a sense of identity and ownership over vector control practices, positioning students as change agents who influence community-wide behaviours. This creates a ripple effect, where knowledge and practices extend beyond schools to families and neighbourhoods, amplifying the programme's impact. Embedding CoP principles enhances intervention sustainability and scalability, reinforcing long-term engagement beyond structured sessions and informing future school-based health initiatives.

Multiple theoretical frameworks are employed in the study to establish a structured methodology for intervention design, implementation and evaluation. CFIR 2.0 provides a robust guiding structure to systematically assess implementation determinants from study inception to evaluation.⁴³ It not only enables a systematic identification of barriers and facilitators to implementation but also aids in developing a structured logic model to visualise intervention pathways linking intervention components with expected outcomes. COMBI strategies are used in fostering targeted communication and social mobilisation (WHO, 2012). Meanwhile, BCW will be instrumental in formulating tailored interventions by systematically analysing the multiple levels of influence like capability, opportunity and motivation.⁴⁹ The RE-AIM framework will facilitate a structured comprehensive evaluation by measuring the programme implementation and effectiveness.⁴⁷ Leveraging multiple frameworks is expected to enhance methodological rigour, provide a comprehensive understanding of intervention dynamics and allow for tailored strategies addressing both structural and behavioural determinants. However, the need for contextual adaptations may complicate implementation and hinder the synthesis of findings across studies.⁵⁷

The study is expected to generate critical evidence on the effectiveness and feasibility of a school-based

intervention for dengue control in India. School-based programmes have shown to be effective in fostering health-related behavioural change, leveraging structured learning environments to instil long-term preventive practices.⁵⁸ By leveraging structured education, experiential learning and peer influence, the intervention aims to extend dengue prevention practices beyond schools to families and communities. Moreover, the findings could inform policy integration within School Health and Wellness Programme under the Ministry of Health and Family Welfare⁵⁹ reinforcing the role of behaviour change communication in vector control strategies. Additionally, embedding behavioural change for vector control from an early age is expected to enhance cleanliness practices and encourage proper solid waste management improving overall public health outcomes.⁶⁰

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Contributors ArR led the conceptualization and design of the DREAMS' 24 project protocol and drafted the initial version of the manuscript. SS contributed to the implementation framework and development of coordination strategy with schools. JP contributed to the identification and application of relevant implementation theories across the project phases and also assisted in workflow visualization in the manuscript. GV assisted in designing the study in line with intersectoral collaboration influencing implementation. AmR formulated the data analysis plan and was a major contributor in writing the manuscript. ANS framed strategies for entomological investigations and provided critical inputs to the manuscript. VKR provided technical guidance on the public health and surveillance components of the protocol. ST provided technical guidance on the educational implementation aspects pertaining to school children and teachers. MR provided overall supervision, critical review of the manuscript, and final approval for submission. All authors reviewed and approved the final manuscript and agreed to be accountable for all aspects of the work. ArR acted as guarantor and is responsible for the overall content of the study. All authors read, reviewed and approved the final manuscript. Napkin AI tool was used to create a visually engaging workflow of the project in Figure 1: Key Activities proposed under DREAMS.

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ORCID iDs

Arya Rahul <https://orcid.org/0000-0003-2383-8482>

Jency Priskilla <https://orcid.org/0009-0004-8889-579X>

Manju Rahi <https://orcid.org/0000-0003-0932-0935>

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